# Identity as Event History: A Worked Introduction to Spherepop

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**Abstract**

Spherepop is a process-oriented formalism in which the identity of an object is determined not by its present configuration but by the irreversible sequence of transformations that produced it. This essay develops the central principles of the framework through a worked example and then situates those principles within increasingly general mathematical structures: event words and normal forms, directed acyclic event graphs, a categorical semantics of histories as functors from causal index categories, a rewriting system for normalization, and connections to trace theory, Petri nets, and string diagram formalisms. Additional sections develop the order-theoretic structure of event histories as posets and their linear extensions, a small-step operational semantics for histories as labeled transition systems, measures of historical complexity including depth and width, the historical reconstruction problem and its relationship to informational irreversibility, a comparison of historical and structural identity criteria, a treatment of Spherepop as a free traced monoidal construction, and the connection to provenance graphs and distributed event logs. Throughout, irreversibility is treated as the fundamental feature that gives event histories their directed, acyclic character and that grounds a coherent alternative to set-membership-based ontology.

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# 1 Introduction

A recurring question in the foundations of mathematics and formal ontology concerns the correct criterion of identity for objects that exist within processes rather than in isolation. Classical set theory locates identity in the membership relation: two sets are equal precisely when they contain exactly the same elements, irrespective of how those sets came to exist or in what order their elements were assembled. This extensional criterion is powerful and convenient, but it discards what may be the most significant feature of many objects encountered in physics, computation, and the theory of complex systems: namely, that an object's identity is inseparable from the history that generated it.

Spherepop is a formalism developed to address this gap directly. Its central claim is that objects should be represented not as static containers but as *records of transformations*, and that identity should be adjudicated by comparing those records rather than by comparing present states. The formalism takes irreversibility as its organising principle. Because the transformations that produce objects cannot in general be undone, the historical record accumulates monotonically; each new event appends to a trace that cannot be retracted. Two objects are therefore identical in the Spherepop sense if and only if they are the products of identical histories.

This orientation connects Spherepop to a broad family of process-based theories. In category theory, the importance of morphisms over objects is a foundational methodological commitment [Mac Lane, 1998]; histories composed of irreversible events are naturally interpreted as compositions of morphisms, and the event histories of objects become diagrams in causal categories. In theoretical computer science, the notion that computational identity should be grounded in execution traces rather than in present memory states appears implicitly throughout operational semantics [Abramsky and Jung, 1994]. In concurrency theory, Mazurkiewicz traces [Mazurkiewicz, 1987] provide an established formalism for treating event sequences as equivalence classes modulo independent commutation, and Spherepop event words can be understood as canonical representatives of such classes. In the theory of concurrent processes, Petri nets [Petri, 1966] provide a complementary operational picture.

The present essay proceeds through several layers of increasing abstraction. Section 2 presents the primary worked example, introducing two processes that produce structurally similar objects by distinct histories and establishing the foundational identity criterion. Section 3 formalises this criterion through the notions of event words and normal forms. Section 4 develops the categorical interpretation, treating events as morphisms and histories as compositions. Section 5 examines the relationship between irreversibility and entropy. Section 6 develops the directed acyclic graph representation of event histories. Section 7 introduces normalization and canonical representations. Section 8 provides the functorial perspective. Section 9 examines merge events as pushout-like constructions. Section 10 interprets histories as trajectories across an entropy landscape. Section 11 develops the partial-order structure of event histories, establishing that event words are linear extensions of the causal precedence relation.

Section 12 introduces quantitative measures of historical complexity including depth, width, and branching factor. Section 13 provides a small-step operational semantics in which histories arise as execution traces of labeled transition systems. Section 14 examines the historical reconstruction problem and its relationship to informational irreversibility. Section 15 compares the Spherepop historical identity criterion with the extensional criterion of classical set theory. Section 16 develops a stratified family of weaker equivalence relations on histories. Section 17 situates the framework within the theory of free traced monoidal categories. Section 18 connects Spherepop to provenance graphs, distributed event logs, and blockchain-style ledgers. Appendices treat the BNF grammar for Spherepop expressions, the rewriting system for normalization, the relationship to trace theory, Petri nets, and string diagrams, and derived causal categories.

## 2 A Worked Example

To make the core principle concrete, consider an initial region $R$ and a sequence of three events. The first event, $E_1$, causes $R$ to split into two descendant regions $A$ and $B$. The second event, $E_2$, is a transformation internal to $A$ that produces a new region $C$. The third event, $E_3$, causes the regions $B$ and $C$ to merge, producing the region $D$. Although the informal description is clear, the formal representation of this history must record not merely which events occurred but in which order they occurred and from which states each event proceeded.

Using the normal-form notation to be fully developed in Section 3, the history of each region is recorded as follows. Both $A$ and $B$ arose directly from $R$ through the single event $E_1$, so their normal forms are identical:

$$(\text{spherepop}(A)) = (E_1), \qquad (\text{spherepop}(B)) = (E_1).$$

The region $C$, however, arose from $A$ through the subsequent event $E_2$, so its history includes both events:

$$(\text{spherepop}(C)) = (E_1, E_2).$$

Finally, $D$ arose through the merger of $B$ and $C$, which itself required all three events in order, so:

$$(\text{spherepop}(D)) = (E_1, E_2, E_3).$$

The identity of each object is thus entirely determined by the ordered sequence of events that produced it. This is already a departure from structural or compositional criteria: a criterion based on present structure would describe $D$ solely in terms of what $D$ currently contains, with no record of whether $D$ arose from a merger of previously split regions or from some entirely different sequence of transformations.

To make this distinction sharp, consider a second process. Suppose again that $E_1$ causes $R$ to split into $A$ and $B$, but now suppose that instead of $E_2$ there occurs a different event $E_3'$ that directly merges $A$ and $B$ into a region $D'$, without the

intermediate transformation that in the first process generated $C$. The normal form of $D'$ is then:

$$(\text{spherepop}(D')) = (E_1, E_3').$$

Comparing the normal forms of $D$ and $D'$,

$$(\text{spherepop}(D)) = (E_1, E_2, E_3), \qquad (\text{spherepop}(D')) = (E_1, E_3'),$$

the two histories are evidently distinct: $D$ accumulated three events while $D'$ accumulated only two, and the events themselves differ. The Spherepop identity criterion therefore concludes that $D$ and $D'$ are not identical objects, even if their present structural descriptions happen to agree.

This is the core claim of the formalism. Structural indistinguishability is insufficient for identity. Two objects are identical in Spherepop precisely when their complete generative histories, recorded in normal form, are identical.

# 3 Event Words and Normal Forms

The worked example of Section 2 illustrates the central representational idea intuitively. To deploy it rigorously, the notion of a history must be formalized as a structured symbolic object.

Let $\mathcal{E}$ denote the set of possible events in a given system. Each event in $\mathcal{E}$ corresponds to an irreversible transformation that either produces new regions, modifies existing ones, or both. The precise physical or computational content of these events may vary; what matters for the formalism is that they are discrete, ordered, and irreversible.

**Definition 1** (Event word)**.** *An* event word *over $\mathcal{E}$ is a finite ordered sequence*

$$w = (E_{i_1}, E_{i_2}, \ldots, E_{i_k})$$

*where each $E_{i_j} \in \mathcal{E}$. The length of $w$ is $|w| = k$. The empty event word, denoting an object with no generative history, is written $\varepsilon$.*

Because the events constituting a word are ordered, the same multiset of events occurring in different sequences represents a different history. This ordering is not arbitrary: it encodes the causal and temporal precedence relations among the events, about which more will be said when branching histories are considered.

**Definition 2** (Spherepop normal form)**.** *Spherepop associates to every object $X$ a canonical expression, the* normal form *of $X$, written $(\mathit{spherepop}(X))$, which is defined to be the event word representing the complete generative history of $X$:*

$$(\mathit{spherepop}(X)) = (E_{i_1}, E_{i_2}, \ldots, E_{i_k}).$$

The normal form is the primary identity criterion of the formalism. Two objects are identical if and only if their normal forms coincide:

$$X \equiv Y \quad \Longleftrightarrow \quad (\text{spherepop}(X)) = (\text{spherepop}(Y)).$$

This principle replaces identity criteria grounded in structural or compositional properties. Objects that are structurally indistinguishable may nevertheless be distinct if their histories differ, and objects that arose through entirely different structural pathways may be identical if those pathways produced identical event sequences.

It is useful to visualize this criterion geometrically. Figure 1 represents an event word as a directed trace along a time axis, with the irreversible events $E_1$, $E_2$, $E_3$ appearing as marked transitions separating successive phases of the history. The tick marks crossing the trace axis emphasize that each transition is a genuine discontinuity: no reversion to a prior state is possible. The brace beneath the axis indicates that the normal form $(\text{spherepop}(D))_{\text{nf}} = (E_1, E_2, E_3)$ records the complete ordered content of that trace.

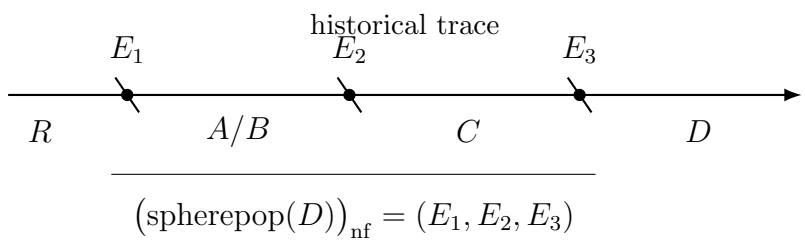


Figure 1: A geometric representation of an event word as an irreversible trace. Tick marks indicate irreversible transitions; the brace records the corresponding normal form.

The geometric picture reinforces a fundamental intuition: an object's position on this trace cannot be recovered by inspecting the object alone. Just as a physical process that has dissipated energy cannot be run backwards to recover the initial configuration, a Spherepop object does not carry within itself the information necessary to reconstruct its generating history. The normal form is an external record, not an intrinsic property.

This historical criterion avoids several classical difficulties of set-based ontology. Objects in Spherepop are not defined by membership in collections but by the irreversible sequences that produced them. The ontology is therefore inherently process-based: objects are histories rather than containers, and the content of an object is its temporal structure rather than its spatial extension.

# 4 A Categorical Interpretation of Spherepop Histories

The representation of objects as event words suggests an immediate connection with category theory, where the primacy of morphisms over objects is a foundational methodological principle [Mac Lane, 1998]. In the categorical framework, processes rather than states are primary; objects emerge as the sources and targets of transformations, and the composition of morphisms encodes the sequential structure of processes. This perspective aligns precisely with the Spherepop principle that identity is grounded in generative history.

Let $\mathcal{C}$ be a category whose objects represent regions or configurations of a system and whose morphisms represent irreversible events. An event

$$E : X \to Y$$

corresponds to a transformation that produces the region $Y$ from the region $X$. If two events occur sequentially,

$$E_1 : X \to Y, \qquad E_2 : Y \to Z,$$

their categorical composition

$$E_2 \circ E_1 : X \to Z$$

represents the combined historical process producing $Z$ from $X$. In this formulation, the event word introduced in Section 3 corresponds precisely to a composition of morphisms, and the normal form of an object is the morphism that takes the initial state to the present one.

Branching and merging events, which are prominent features of the worked example, correspond to familiar categorical constructions. A split event

$$E_1 : R \to A \sqcup B$$

resembles a coproduct-like branching in which a single region gives rise to multiple descendants, while a merge event

$$E_3 : (B, C) \to D$$

resembles a pushout-like construction in which two regions produced from a common ancestor are combined into a new one [Mac Lane, 1998, Spivak, 2014]. The precise categorical status of these constructions is examined further in Section 9; for present purposes it suffices to note that the full history of the worked example forms a diagram in $\mathcal{C}$ of the shape illustrated in Figure 2.

From this perspective, the identity of an object is determined by the entire diagram of morphisms that generated it. Two objects are identical precisely when the diagrams representing their histories are identical, up to equality of morphisms. This is a stronger

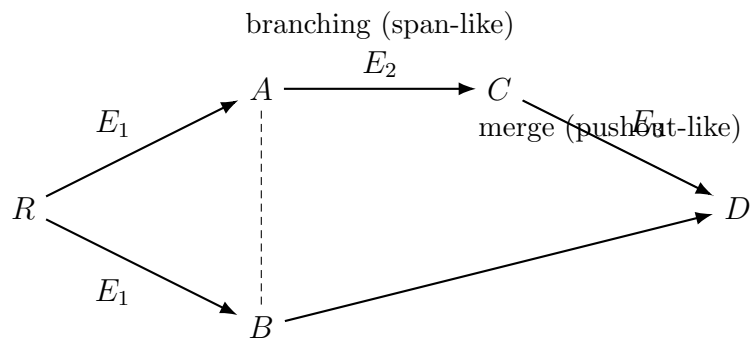


Figure 2: A categorical diagram of the worked example. The split event $E_1$ produces a span, while the merge event $E_3$ completes a pushout-like construction. The identity of $D$ is determined by the entire diagram, not merely by the node $D$ in isolation.

condition than structural isomorphism of the resulting objects; it requires agreement at every stage of the generative process.

The categorical viewpoint also clarifies why Spherepop naturally produces directed acyclic event graphs. Cycles in a morphism-composition diagram would correspond to reversible processes, since a path

$$X \to Y \to \cdots \to X$$

implies the existence of a composite morphism returning from $X$ to itself. In a setting where morphisms represent irreversible transformations, such a cycle would violate the irreversibility condition. The absence of cycles is therefore not merely a convenient property but a structural consequence of the physical or computational assumption that the events under consideration are irreversible. The relationship between irreversibility and directed acyclicity is developed further in Section 6, and the connection to entropy is examined in Section 5.

Baez and Stay [2010] describe in detail the deep correspondences among physics, topology, logic, and computation through the "Rosetta Stone" of categorical structures; the Spherepop framework sits naturally within the physical and computational columns of that correspondence, where morphisms encode processes and composition encodes sequential execution.

# 5 Irreversibility and Entropy in Event Histories

The irreversibility of Spherepop events is not merely a formal stipulation; it reflects a substantive claim about the physical and informational character of the processes under consideration. In thermodynamic systems, irreversible transformations correspond to processes in which entropy increases and in which information about prior states becomes dispersed across degrees of freedom that are no longer practically accessible. The connection between irreversibility and entropy is therefore a natural one to explore

within the Spherepop framework, where each event appends a new term to a historical record that cannot be retracted.

Consider again an event $E : X \to Y$ representing a transformation that produces the region $Y$ from the region $X$. In general, the inverse transformation does not exist: one cannot recover $X$ uniquely from $Y$ alone. Consequently the history of $Y$ cannot be reconstructed without additional information beyond what $Y$ currently contains. Spherepop responds to this observation by treating the event history itself as the primary record, so that the normal form

$$(\text{spherepop}(X)) = (E_{i_1}, E_{i_2}, \ldots, E_{i_k})$$

functions as a symbolic trace of the entropy-producing sequence of operations that generated $X$. Each event appends a new irreversible step to this trace, and the trace cannot be shortened by any transformation available within the system.

This motivates a natural measure of historical complexity. If $H(X)$ denotes the length of the event word associated with $X$,

$$H(X) = |(\text{spherepop}(X))|,$$

then $H(X)$ provides a simple count of the irreversible operations that separates $X$ from its initial state. Each additional event increases the informational content required to specify the object's origin.

More sophisticated formulations may weight individual events by their informational or thermodynamic significance. Define a weight function $w : \mathcal{E} \to \mathbb{R}_{\geq 0}$ assigning to each event $E$ a non-negative real number $w(E)$ measuring its entropy contribution. The weighted historical complexity of $X$ is then

$$S(X) = \sum_{j=1}^{k} w(E_{i_j}).$$

When $w$ is chosen to reflect genuine thermodynamic quantities, $S(X)$ provides an estimate of the total entropy produced along the history of $X$. In general, the choice of $w$ encodes assumptions about the relative significance of different classes of events, and different physical or computational models will motivate different choices.

The geometric interpretation of this measure is illustrated in Figure 3. As events accumulate, the historical trace widens: the cross-sectional thickness of the trajectory at each stage reflects the growing informational content required to describe the object's origin. Critical events that significantly increase $S(X)$ correspond to points of rapid widening, while low-entropy events produce only modest expansion.

This interpretation connects Spherepop to a broader class of process-based theories in which the fundamental objects of a theory are not static states but trajectories through configuration space. In such theories, the informational content of an object is the information required to specify its trajectory rather than merely its present

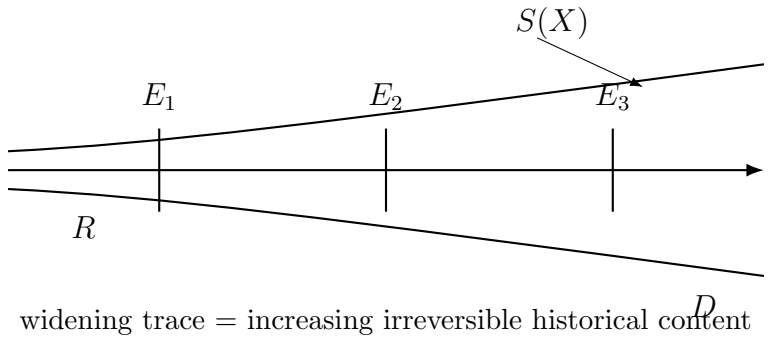


Figure 3: Irreversible history represented as geometric widening. As events accumulate, the historical trace broadens, representing the growing informational content that would be required to reconstruct the prior state. The function $S(X)$ measures cumulative weighted entropy along the trace.

position. The Spherepop normal form functions as a compact symbolic encoding of this trajectory.

The resulting ontology is one in which entropy production provides the arrow of time. Because $S(X)$ is non-decreasing along any history—since all events are assumed to contribute non-negatively to the weighted sum—histories are naturally directed. This directionality is not imposed externally as a formal condition but emerges from the physical assumption that irreversible processes increase entropy. The directed acyclic structure of event histories, which is treated as a formal property in the following section, therefore has a thermodynamic grounding in the behaviour of $S(X)$ along any path through the event graph.

# 6 Event Graphs and Directed Acyclic Structure

Linear event words provide a correct representation of histories in which each event has at most one predecessor and at most one successor. The processes encountered in practice, however, frequently involve branching and merging: a single region may give rise to multiple descendants, and multiple regions may combine to produce a single new one. In such cases the linear notation, though applicable after a suitable ordering is imposed, obscures the parallel structure of the history. The natural representation is a directed graph.

**Definition 3** (Event graph)**.** *An* event graph *is a directed graph $G = (V, \mathcal{E})$ in which the vertices $V$ represent regions and each directed edge $X \to Y$ represents an irreversible event producing the region $Y$ from the region $X$.*

When events produce multiple descendants or require multiple inputs, the corresponding vertices in $G$ carry multiple outgoing or incoming edges respectively. The

complete generative history of a system is encoded in the global structure of $G$ rather than in any single path through it.

**Proposition 1** (Directed acyclicity)**.** *Every Spherepop event graph is a directed acyclic graph (DAG).*

*Proof.* Suppose for contradiction that $G$ contains a directed cycle $X \to Y \to \cdots \to X$. Then there exists a composite event morphism that maps $X$ to itself, implying that the transformation can be applied repeatedly to yield a reversible process taking $X$ back to a state indistinguishable from $X$. Since every Spherepop event is irreversible by definition, such a composite cannot exist. Consequently no directed cycle can occur in $G$. □

The directed acyclic property is not merely a convenience; it is the formal analogue of the thermodynamic irreversibility condition formalized in Section 5. Every finite DAG admits a topological ordering, and it is precisely this ordering that will be exploited by the normalization procedure of Section 7 to produce canonical event words from branching histories.

Given an object $X$ appearing as a vertex in $G$, its identity is determined not by the local properties of the vertex but by the global structure of the subgraph that generated it. Formally, let $\mathrm{Anc}(X)$ denote the set of all vertices from which $X$ is reachable by a directed path in $G$. The *ancestral subgraph* of $X$ is the induced subgraph

$$G_X = G[\mathrm{Anc}(X) \cup \{X\}]$$

containing all vertices that contributed, directly or indirectly, to the production of $X$.

**Definition 4** (Spherepop identity, graphical formulation)**.** *Two objects $X$ and $Y$ are Spherepop-identical if and only if their ancestral subgraphs $G_X$ and $G_Y$ are isomorphic as directed graphs with edges labelled by event types.*

This definition subsumes the event-word formulation of Section 3 as a special case: when histories are linear (no branching or merging), the ancestral subgraph reduces to a directed path, and graph isomorphism reduces to equality of the corresponding event sequences.

Figure 4 illustrates a more complex event graph containing multiple levels of branching. The terminal object $Z$ is determined by the entire layered DAG structure feeding into it. Different subsets of the graph characterise different terminal objects; the identity criterion requires that the complete ancestral subgraph, not merely the immediately preceding events, be taken into account.

The DAG representation also clarifies the relationship between the event-word notation and the diagrammatic structure. A linear event word corresponds to a directed path through the DAG. Branching corresponds to a vertex with multiple outgoing edges, and merging corresponds to a vertex with multiple incoming edges. The graph therefore provides a complete representation of the causal structure, while the event

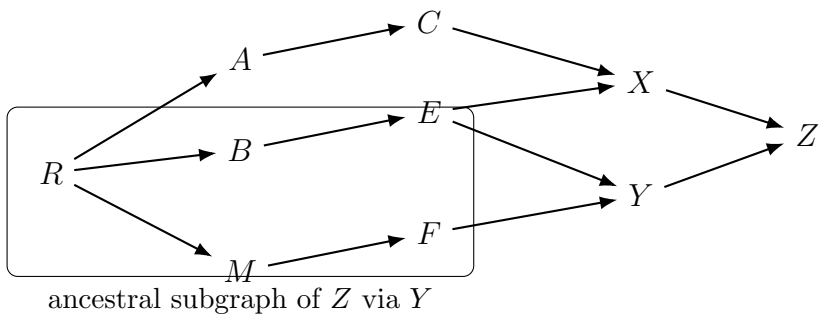


Figure 4: A layered directed acyclic event graph. The identity of the terminal object $Z$ is determined by the entire ancestral subgraph that feeds into it, not merely by its immediate predecessors $X$ and $Y$.

word provides a linear serialization of that structure. The correspondence between these two representations is the subject of the normalization theory in Section 7, where it is shown that every event graph admits a canonical linear serialization.

# 7 Normalization and Canonical Representation

The event graph representation captures the full causal structure of a history, but different textual descriptions of the same graph may arise depending on the order in which causally independent events are recorded. If two events act on disjoint regions without either depending on the output of the other, then listing them in either order produces an equivalent description of the same causal structure. This equivalence must be recognized formally if the identity criterion is to be stable under changes of representation.

**Definition 5** (Event independence)**.** *Two events $E_i, E_j \in \mathcal{E}$ are* independent*, written $E_i \parallel E_j$, if neither event requires the output of the other as input; equivalently, there is no directed path between $E_i$ and $E_j$ in the event graph.*

Independent events may appear in either order in a linear serialization of the event graph without altering the graph's structure. This motivates the following rewriting rule, which will be made precise in Appendix B:

$$(E_i, E_j) \longrightarrow (E_j, E_i) \quad \text{whenever } E_i \parallel E_j.$$

Two event words are *causally equivalent* if they are related by a finite sequence of such commutation steps. The normalization procedure selects a canonical representative from each causal equivalence class.

**Definition 6** (Spherepop normal form, canonical)**.** *The* Spherepop normal form *of an object $X$ is the event word*

$$(\mathit{spherepop}(X))_{\mathrm{nf}}$$

*obtained by computing a topological ordering of the event graph $G_X$ and serializing the events in that order, breaking ties consistently according to a fixed deterministic convention on event labels.*

Because every finite DAG admits a topological ordering, and because the normalization procedure is deterministic, the normal form is well-defined for every Spherepop object. The canonical identity condition becomes:

$$X \equiv Y \quad \Longleftrightarrow \quad (\mathrm{spherepop}(X))_{\mathrm{nf}} = (\mathrm{spherepop}(Y))_{\mathrm{nf}}.$$

Two objects are identical exactly when their normalized event histories coincide, regardless of the particular serialization that may have been used to describe their histories in the first instance.

Figure 5 illustrates this point concretely. Two syntactically different presentations of the same history—one listing $E_i$ before $E_j$ and the other listing $E_j$ before $E_i$ on two parallel branches—both normalize to the same canonical form, because the two events are independent and the topological ordering convention resolves the tie deterministically.

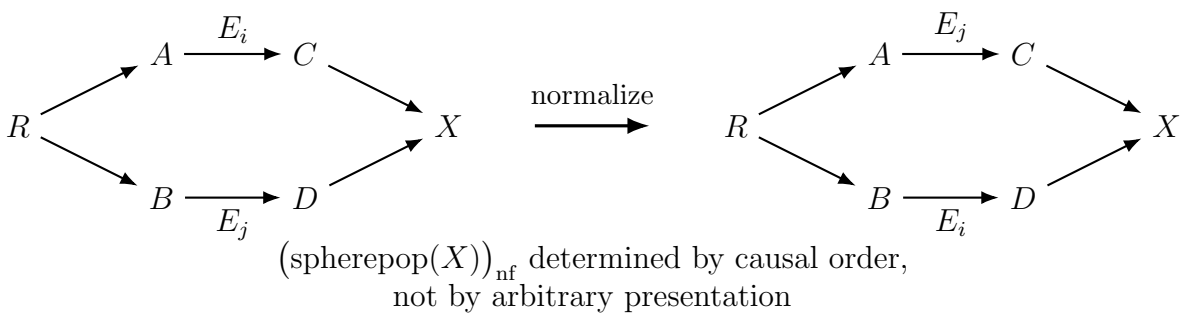


Figure 5: Two syntactically different presentations of independent events $E_i$ and $E_j$ encoding the same causal structure. Normalization collapses both to a single canonical form, since $E_i \parallel E_j$ implies their order may be resolved by any fixed topological convention.

The normalization procedure plays a role in Spherepop analogous to that of canonical forms in other mathematical frameworks: reduced words in group theory, row echelon form in linear algebra, beta-normal forms in the lambda calculus. Each of these devices converts a syntactically diverse family of equivalent expressions into a unique representative that can be compared directly. In Spherepop, the canonical form ensures that identity depends only on causal structure rather than on the order in which an external observer happened to record causally independent events.

# 8 Functorial Perspectives on Event Histories

The event graph representation and the normalization theory of Section 7 can be cast in a fully categorical language by treating the process of mapping abstract causal structures

onto concrete systems as a functor. This perspective, which draws on the categorical foundations developed in Mac Lane [1998] and elaborated in Spivak [2014] and Baez and Stay [2010], provides the most general formulation of Spherepop semantics.

**Definition 7** (Causal history category)**.** *Let $\mathcal{H}$ be a small category whose objects represent intermediate regions or states in a history and whose morphisms represent irreversible events. Because the events are temporally ordered and irreversible, if there exists a morphism $X \to Y$ in $\mathcal{H}$ then there cannot exist a morphism $Y \to X$. Such a category is called a* causal history category*.*

In practice, $\mathcal{H}$ may be constructed directly from the event graph $G$ by interpreting vertices as objects and directed edges as generating morphisms, with composition given by directed paths. The temporal ordering that the DAG structure imposes on $G$ corresponds to the "no-back-morphism" condition in $\mathcal{H}$.

**Definition 8** (Spherepop realization functor)**.** *Let $\mathcal{C}$ be a category of regions, configurations, or system states. A* Spherepop realization *of the history $\mathcal{H}$ in $\mathcal{C}$ is a functor*

$$F : \mathcal{H} \to \mathcal{C}.$$

The functor $F$ assigns to each node in the abstract causal history a concrete object in $\mathcal{C}$, and to each event morphism the corresponding transformation in the physical or computational system. Functoriality ensures that sequential composition of events in $\mathcal{H}$ corresponds exactly to sequential composition of transformations in $\mathcal{C}$:

$$F(E_2 \circ E_1) = F(E_2) \circ F(E_1).$$

This coherence condition is the categorical expression of the physical requirement that the combined effect of two sequential events is the composition of their individual effects, without hidden intermediate states.

Terminal objects of $\mathcal{H}$ correspond to the currently existing objects of the system—those that have not been further transformed. If $X$ is such a terminal node, then $F(X)$ represents the concrete present state produced by the entire chain of events that preceded it. The identity of $F(X)$ as a Spherepop object is determined jointly by $F$ and by the position of $X$ in the causal diagram.

Figure 6 illustrates the functorial picture. The abstract causal history category $\mathcal{H}$ on the left carries the relational structure of the history—which events preceded which—without specifying their concrete content. The functor $F$ maps this abstract structure into the category $\mathcal{C}$ of realized regions on the right, where each abstract node becomes a concrete object and each abstract morphism becomes a concrete transformation.

The functorial formulation also provides a clean account of when two objects should be considered identical. Suppose

$$F_1 : \mathcal{H}_1 \to \mathcal{C} \quad \text{and} \quad F_2 : \mathcal{H}_2 \to \mathcal{C}$$

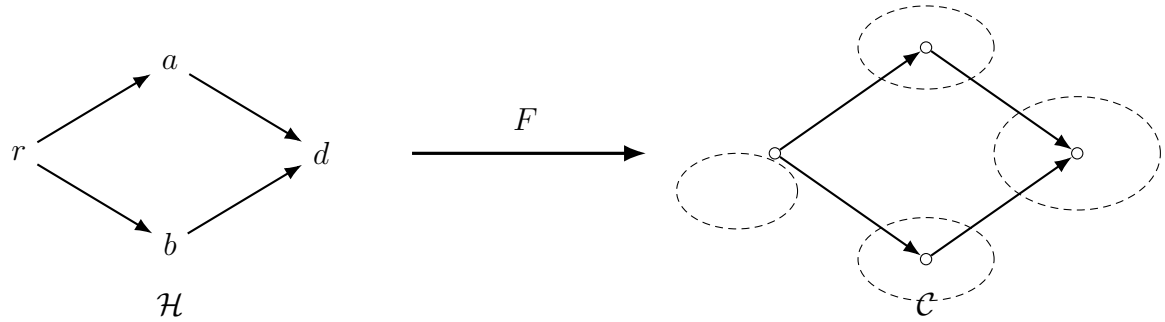


Figure 6: A Spherepop history as a functor $F : \mathcal{H} \to \mathcal{C}$ from an abstract causal index category $\mathcal{H}$ to a category $\mathcal{C}$ of realized regions. The abstract structure on the left encodes causal precedence; the functor translates this into concrete transformations on the right.

are two realizations. The resulting terminal objects coincide as Spherepop objects when the normalized causal structures $\mathcal{H}_1$ and $\mathcal{H}_2$ are isomorphic as causal categories and the functors agree on the corresponding morphisms under this isomorphism. This condition recovers the normal-form identity criterion of Section 7 in categorical language: it requires agreement not merely on terminal values but on the entire structure of the generating process.

The functorial perspective also suggests how Spherepop should handle abstraction. Natural transformations between realizations—that is, morphisms in the functor category $[\mathcal{H}, \mathcal{C}]$—correspond to systematic reinterpretations of a history that change the concrete objects assigned to nodes while preserving the relational structure. In this way the framework supports a hierarchy of increasingly abstract descriptions of the same underlying causal history.

# 9 Merge Events as Pushouts

The merge events that appear prominently in Spherepop histories admit a precise categorical characterization as pushout constructions. A pushout in a category $\mathcal{C}$ is the universal solution to the problem of combining two morphisms that share a common source: given morphisms $f : R \to B$ and $g : R \to C$, the pushout is an object $D$ together with morphisms $B \to D$ and $C \to D$ such that the resulting square commutes and $D$ is universal with this property [Mac Lane, 1998].

The connection to Spherepop is direct. In the worked example of Section 2, the regions $B$ and $C$ both originated from the common ancestor $R$ via the split event $E_1$. The merge event $E_3$ then combined $B$ and $C$ into the region $D$. Categorically, this

situation is described by the commuting square

$$\begin{tikzcd} R \arrow[r] \arrow[d] & C \arrow[d] \\ B \arrow[r] & D \end{tikzcd}$$

in which $D$ receives consistent morphisms from $B$ and $C$ and is universal with this property. The merge event $E_3$ corresponds precisely to the completion of this pushout diagram.

$$\begin{tikzcd} & C \arrow[dr] & \\ R \arrow[ur] \arrow[dr] & \square & D \\ & B \arrow[ur] & \end{tikzcd}$$

Figure 7: A merge event represented as a pushout-like square. The common ancestor $R$ gives rise to $B$ and $C$ via the split event; the merge event $E_3$ completes the square by producing the universal object $D$.

The pushout interpretation makes precise the sense in which the identity of $D$ depends on the entire commuting square rather than merely on the pair $(B, C)$ considered in isolation. Two regions $B'$ and $C'$ could be merged to produce a region $D'$ with the same structural description as $D$, yet if $B'$ and $C'$ arose from a different common ancestor $R'$, the resulting pushout square would be different and $D'$ would be a different Spherepop object. The pushout construction thus encodes the relational history of the merge event, not merely its output.

This observation connects to the broader theme of the paper. In standard set-theoretic or structural accounts of identity, two sets are equal when they contain the same elements, regardless of how those elements were assembled. The pushout characterization of merge events shows that in Spherepop the corresponding condition is far stronger: the merged object must have arisen from the same causal configuration, with the same common ancestor and the same morphisms connecting it to its predecessors.

The pushout interpretation also interacts with the categorical coherence conditions noted in Section 8. A realization functor $F : \mathcal{H} \to \mathcal{C}$ that preserves pushouts maps each merge event in the abstract history to a genuine pushout in the target category. When $\mathcal{C}$ is a category with all pushouts—for instance, a category of sets or a category of topological spaces—this preservation condition is automatically satisfied. In physical applications, the relevant category may not have all pushouts, and the question of when merge events can be faithfully realized is a substantive one.

# 10 Entropy Landscapes and Historical Trajectories

A complementary geometric interpretation of Spherepop histories arises from the notion of an entropy landscape. This interpretation is more impressionistic than the category-theoretic treatment of preceding sections, but it connects the formalism to physical intuitions about the role of irreversibility in shaping the space of possible histories.

Let the configuration space of a system be represented as a manifold $\mathcal{M}$ whose points correspond to possible structural states. Define an entropy-like functional

$$S : \mathcal{M} \to \mathbb{R}$$

measuring the informational or thermodynamic complexity of each state. Irreversible events may then be interpreted as transitions that move the system along trajectories in $\mathcal{M}$ while generally increasing the value of $S$. The history of an object corresponds to a directed path through this landscape, constrained to move in the direction of increasing $S$.

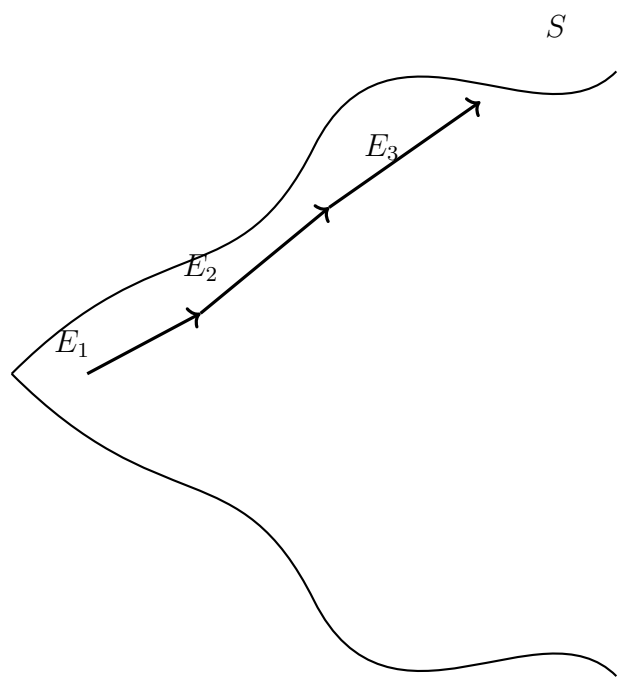


Figure 8: Event history interpreted as a directed trajectory across an entropy landscape. Each event corresponds to a transition along a path that generally moves toward regions of higher entropy. The topology of the landscape constrains which transitions are possible.

In this geometric picture, as illustrated in Figure 8, the events $E_1$, $E_2$, $E_3$ correspond to successive steps along a trajectory that climbs the entropy surface. The shape of

the landscape determines which transitions are available at each point: a configuration at a local minimum of $S$ that is not a global minimum may be accessible from some prior states but inaccessible from others, and the question of reachability depends on the full path taken up to that point. This path-dependence is precisely the feature that Spherepop captures through its event-word representation: the normal form records not just the endpoint of the trajectory but the entire path.

Critical points of the entropy functional may be interpreted as stable structural configurations or attractor states. The Spherepop history of an object records the sequence of transitions that led to its current position on the landscape, including which saddle points and ridges it passed through on the way. Two objects at the same configuration on the landscape but arriving via different paths are therefore different Spherepop objects, even though a purely spatial description would identify them.

The entropy landscape interpretation also suggests a connection to the theory of gradient flows and Morse theory, where the topology of a functional landscape constrains the structure of trajectories and the classification of critical points. Spherepop histories can in principle be studied using these geometric tools to understand, for instance, which pairs of histories are deformable into one another through the insertion or deletion of null events, or how the topology of the causal DAG interacts with the geometry of the underlying landscape. This direction of generalization lies beyond the scope of the present essay but suggests a rich programme of further investigation.

# 11 Event Histories as Partial Orders

The directed acyclic structure of Spherepop event graphs implies that each history carries a natural partial order on its events. Making this structure explicit provides a useful bridge between the graph-theoretic and algebraic representations of histories developed in previous sections, and connects the Spherepop framework directly to classical order theory and the concurrency- theoretic tradition of Mazurkiewicz [Mazurkiewicz, 1987].

Let $G = (V, \mathcal{E})$ be an event graph. Define a binary relation $\prec$ on the set of events by declaring

$$E_i \prec E_j$$

whenever there exists a directed path from $E_i$ to $E_j$ in $G$. The following proposition is immediate from the DAG property established in Section 6.

**Proposition 2.** *The relation $\prec$ is a strict partial order on the set of events of any Spherepop history.*

*Proof.* Irreflexivity holds because the event graph is acyclic: no event can appear on a directed path that begins and ends at itself. Asymmetry is immediate from the same acyclicity: if $E_i \prec E_j$ then there is a directed path from $E_i$ to $E_j$, and the existence of a directed path from $E_j$ to $E_i$ would close a cycle. Transitivity follows by concatenating directed paths: if $E_i \prec E_j$ and $E_j \prec E_k$ then the composed path witnesses $E_i \prec E_k$. $\square$

Each Spherepop history therefore determines a partially ordered set $(\mathcal{E}_X, \prec)$, called the *causal poset* of $X$. The incomparable elements of this poset are precisely the mutually independent events: two events $E_i$ and $E_j$ satisfy $E_i \parallel E_j$ if and only if neither $E_i \prec E_j$ nor $E_j \prec E_i$.

The event words introduced in Section 3 can now be interpreted with precision as *linear extensions* of the causal poset.

**Definition 9** (Linear extension)**.** *A* linear extension *of* $(\mathcal{E}_X, \prec)$ *is a total ordering* $\leq$ *of* $\mathcal{E}_X$ *such that* $E_i \prec E_j$ *implies* $E_i \leq E_j$*.*

Every topological ordering of the event graph $G_X$ corresponds to a linear extension of the causal poset, and conversely every linear extension arises from some topological ordering. The collection of all linear extensions therefore coincides precisely with the set of all event words that are causally equivalent in the sense of Section 7: they represent different serializations of the same partial order.

This observation reframes normalization in order-theoretic terms. The commutation rules that identify independent events are precisely the moves that relate different linear extensions of the same poset. The Spherepop normal form selects a canonical linear extension by fixing a deterministic rule for breaking incomparability ties.

The width of the causal poset, which counts the size of the largest antichain, measures the maximal degree of parallelism in the history. Dilworth's theorem [Mac Lane, 1998] states that the minimum number of chains needed to partition the poset equals the width; in the Spherepop context this means that the most compressed sequential description of a history requires at least as many serialized chains as the width of its causal poset. Wide histories represent highly parallel processes that cannot be faithfully compressed into a single short sequential trace.

The partial-order perspective thus provides the algebraic underpinning for both the normalization theory and the complexity measures developed in the next section, tying these constructions to the classical theory of posets and their linear extensions.

# 12 Historical Complexity Measures

The entropy measure $S(X)$ introduced in Section 5 captures the cumulative informational weight of a history by summing the contributions of individual events. A complementary family of measures can be derived directly from the structure of the causal poset $(\mathcal{E}_X, \prec)$ introduced in Section 11. These graph-theoretic quantities characterize the structural complexity of a history independently of any specific thermodynamic or informational interpretation.

**Definition 10** (Historical depth)**.** *The* depth *of an object* $X$ *is the length of the longest directed path in its ancestral event graph:*

$$\mathrm{depth}(X) = \max\{\, |p| \mid p \textit{ is a directed path in } G_X \,\}.$$

Depth measures the maximal sequential chain of irreversible events separating $X$ from its ultimate ancestor. In a purely linear history depth coincides with the length of the event word, but in branching histories depth may be significantly smaller than the total event count, since many events may proceed in parallel.

**Definition 11** (Historical width)**.** *The* width *of the history of $X$ is the size of the largest antichain in the causal poset:*

$$\mathrm{width}(X) = \max\{\, |A| \mid A \subseteq \mathcal{E}(G_X) \text{ is an antichain}\}.$$

Width measures the maximal degree of concurrent activity present in the history. A history with large width contains many independent events that could in principle occur simultaneously; a history with width one is entirely sequential.

**Definition 12** (Branching factor)**.** *The* branching factor *of the history of $X$ is the maximum number of outgoing edges from any vertex in $G_X$.*

Branching events correspond to moments at which the system diversifies into multiple descendant configurations. A high branching factor indicates that individual transformations generate large numbers of descendants, producing rapidly widening event trees.

These three quantities interact in informative ways. By Dilworth's theorem, as noted in Section 11, the minimum number of chains required to cover the causal poset equals the width; consequently $\mathrm{depth}(X) \cdot \mathrm{width}(X)$ provides a lower bound on the total number of events in the history. Conversely, histories in which depth and width are both small are in a precise sense efficiently compressed: they achieve their complexity through neither excessive sequentiality nor excessive concurrency.

Beyond these three measures one may define finer invariants. The *entropy growth rate* of a history is the ratio $S(X)/\mathrm{depth}(X)$, measuring the average thermodynamic cost per sequential step. The *causal density*, defined as the ratio of the number of edges to the number of vertices in $G_X$, captures how heavily constrained the event graph is by precedence relations. Histories with high causal density are those in which most events depend on most prior events; histories with low causal density are those with sparse causal dependencies and therefore high parallelism.

Together, the complexity measures of this section provide a basic vocabulary for comparing the structural profiles of different generative processes. They complement the entropy functional by describing not the thermodynamic cost of a history but its combinatorial shape—whether it is deep or shallow, narrow or wide, densely or sparsely connected. A full complexity theory of Spherepop histories would develop these measures systematically, establishing bounds on their interactions and identifying which profiles are achievable under given constraints on the event signature.

# 13 Operational Semantics of Spherepop

The preceding sections have treated Spherepop histories primarily as static structures: event words, partial orders, directed graphs, and categorical diagrams. It is equally

important to understand the framework dynamically, as an operational system in which configurations evolve through discrete transitions. This section introduces a small-step operational semantics for Spherepop that makes the execution model explicit and connects the historical identity criterion to the standard vocabulary of process calculi and labeled transition systems [Abramsky and Jung, 1994].

Let $\Sigma$ denote the set of possible system configurations. A *labeled transition system* (LTS) for Spherepop consists of a set $\Sigma$ of configurations, a set $\mathcal{E}$ of event labels, and a transition relation

$$\xrightarrow{E} \;\subseteq\; \Sigma \times \mathcal{E} \times \Sigma,$$

written $X \xrightarrow{E} Y$ when the event $E$ transforms configuration $X$ into configuration $Y$. The irreversibility assumption requires that the relation contain no pair of transitions $X \xrightarrow{E} Y$ and $Y \xrightarrow{E'} X$ for any events $E$ and $E'$.

A Spherepop history arises as an execution trace of this LTS. Starting from an initial configuration $X_0$, a sequence of enabled transitions

$$X_0 \xrightarrow{E_1} X_1 \xrightarrow{E_2} X_2 \xrightarrow{E_3} \cdots \xrightarrow{E_k} X_k$$

generates the object $X_k$. The normal form of $X_k$ is the trace of event labels produced by this execution:

$$(\text{spherepop}(X_k)) = (E_1, E_2, \ldots, E_k).$$

This connection between the static normal form and the dynamic execution trace is the operational content of the Spherepop identity criterion: two objects are identical precisely when no execution producing one can be distinguished from any execution producing the other by examining their event-label sequences.

Branching configurations arise when a single configuration $X$ admits multiple simultaneously enabled transitions acting on disjoint parts of the system. In such cases, rather than choosing an order of execution, the transitions may proceed concurrently. The resulting structure is no longer a linear trace but the directed acyclic event graph described in Section 6, whose vertices are intermediate configurations and whose edges are individual transitions. The normalization of Section 7 selects a canonical serialization of this concurrent execution, corresponding to a particular scheduling order for the independent transitions.

This operational picture also clarifies the role of the empty event word $\varepsilon$ introduced in Section 3. In the LTS formulation, $\varepsilon$ corresponds to the identity trace: an object in the initial configuration $X_0$ that has undergone no transitions. Such an object is defined entirely by its starting state, with no generative history to record.

The relationship between the Spherepop operational semantics and standard bisimulation equivalences is instructive. In the theory of process calculi, two processes are bisimilar when each can simulate the other step by step [Abramsky and Jung, 1994]. The Spherepop identity criterion is strictly stronger than bisimilarity: it requires not merely that two objects respond to future transitions in the same way, but that they

arrived at their present state via identical histories. Bisimulation looks forward; Spherepop identity looks backward. This contrast mirrors the broader distinction between structural and historical identity discussed in Section 15.

# 14 The Historical Reconstruction Problem

The operational semantics of Section 13 raises a natural inverse problem. Given an object $X$ in its present configuration, is it possible to recover the history that produced it? In classical Hamiltonian mechanics, reversibility ensures that the present state uniquely determines all past states. Spherepop, by contrast, is grounded in irreversibility. The question of historical reconstruction therefore has a fundamentally different character.

**Definition 13** (Historical reconstruction problem)**.** *Given a present object $X$ with configuration $C$, the* historical reconstruction problem *asks for the set*

$$\mathcal{R}(X) = \{\, H \mid H \textit{ is an event history producing a configuration isomorphic to } C \,\}.$$

In general $\mathcal{R}(X)$ is not a singleton. Multiple distinct histories may converge to structurally indistinguishable configurations, as the worked example of Section 2 illustrates: the objects $D$ and $D'$ may be structurally identical even though their generating histories are distinct. The reconstruction problem therefore typically admits infinitely many solutions, corresponding to the infinitely many event sequences that could in principle have generated the observed configuration.

This indeterminacy is not a defect of the formalism but a reflection of an informational fact about irreversible processes. Each irreversible event dissipates information about the prior configuration, so that the prior state is in general not recoverable from the posterior state alone. The Spherepop normal form resolves this ambiguity by treating the history as a primitive datum: the identity of $X$ is not the configuration $C$ but the specific history $H$ that produced it.

From the perspective of causal inference, the reconstruction problem corresponds to asking which causal structures are consistent with the observed data. In many applications, partial information about the history may be available from side channels—for instance, timestamps, audit logs, or provenance metadata—and the problem reduces to disambiguating among histories consistent with this partial information. The Spherepop framework provides a formal language in which such disambiguation can be stated: one seeks a history $H$ satisfying both the structural constraint $H \in \mathcal{R}(X)$ and any additional constraints imposed by the available side-channel information.

The reconstruction problem also has thermodynamic significance. The number of distinct histories in $\mathcal{R}(X)$ may be interpreted as an entropy of the set of possible pasts compatible with the present observation. A large set $\mathcal{R}(X)$ indicates that the present state is informationally impoverished with respect to the past: many different processes could have generated it. A singleton $\mathcal{R}(X)$ would indicate a maximally informative present state, one from which the unique generating history can be fully recovered—but in an irreversible system this is the exceptional rather than the typical case.

By foregrounding this asymmetry, the Spherepop framework aligns with a broadly held principle in the philosophy of time: the past is determined and the future is open, but the past is often informationally inaccessible from the present. The normal form preserves the information that would otherwise be lost through irreversibility, precisely by treating it as constitutive of identity rather than as a contingent annotation.

# 15 Historical Identity and Structural Identity

The central claim of the Spherepop framework is that identity should be understood historically rather than structurally. It is worth making this contrast precise by examining how the two criteria relate in formal terms, and what each implies for the ontology of objects.

Classical set-theoretic identity is extensional. Two sets are equal when they contain the same elements, regardless of how they were constructed:

$$A = B \quad \Longleftrightarrow \quad \forall x\,(x \in A \leftrightarrow x \in B).$$

This criterion is powerful precisely because it discards all information about the generative process. It allows one to identify any two sets with the same extension, irrespective of whether they were assembled through different sequences of operations or defined by entirely different predicates.

Spherepop replaces this extensional criterion with a historical one. The identity of an object $X$ is determined by the ordered sequence of events that produced it:

$$X = Y \quad \Longleftrightarrow \quad (\text{spherepop}(X))_{\text{nf}} = (\text{spherepop}(Y))_{\text{nf}}.$$

Two objects are identical only when their entire generative histories coincide. This criterion is strictly more discriminating than structural equality: it distinguishes objects whose present configurations agree but whose histories do not, and it identifies objects only when the complete record of their production matches.

The distinction is illustrated in Figure 9. Two distinct trajectories $H_1$ and $H_2$ converge to the same point in configuration space. From a structural perspective the terminal objects are indistinguishable; from the Spherepop perspective they are genuinely distinct because their generating histories differ.

This difference matters in contexts where the history of a transformation carries independent significance. In thermodynamic systems, two states that are thermodynamically equivalent at the present moment may nevertheless have arrived there through processes with different entropy productions; the processes are physically distinct even if the states are macroscopically indistinguishable. In distributed computation, two datasets with identical current contents may have been produced by different workflows; for audit and provenance purposes they should be treated as distinct artifacts. In biological development, two cells with the same gene-expression profile may have followed different differentiation pathways; their developmental histories constrain their future behavior in ways that their present profiles do not capture.

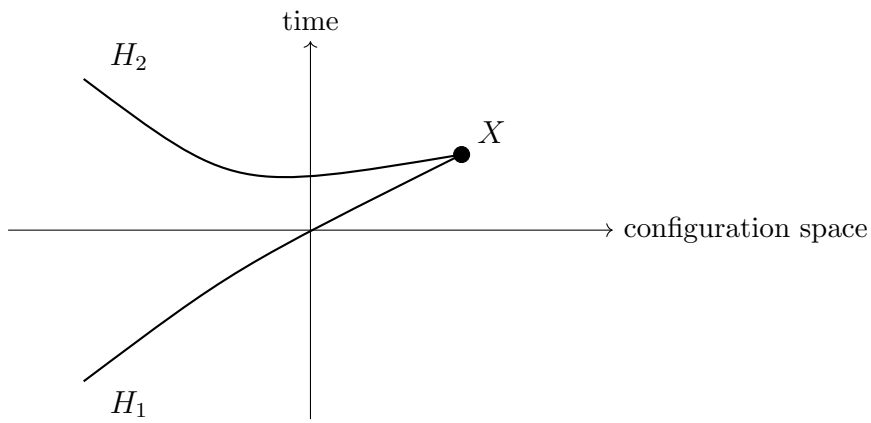


Figure 9: Two histories $H_1$ and $H_2$ converging to the same terminal configuration. A structural criterion identifies the resulting objects; the Spherepop historical criterion distinguishes them.

In each of these cases the Spherepop criterion provides a formalization of a criterion that practitioners already employ implicitly: identity is not merely a matter of present structure but of the history that produced it.

It is also worth noting the relationship between historical identity and the weaker equivalence relations considered in Section 16. The full Spherepop identity criterion lies at the most discriminating end of a spectrum. At the other extreme, structural isomorphism ignores history entirely and identifies any two configurations of the same type. Between these extremes lie equivalences that preserve some but not all historical information. Understanding the lattice of these equivalences clarifies which aspects of history are being retained or discarded under each description.

# 16 Equivalence Relations on Histories

The Spherepop identity criterion identifies two objects precisely when their normalized event histories coincide. This is the strictest reasonable notion of identity within the framework. However, many applications require coarser comparisons—situations in which two objects are considered equivalent even though their detailed histories differ. This section introduces a stratified family of equivalence relations that captures the range of positions between strict historical identity and pure structural indistinguishability.

The finest relation is Spherepop identity itself, which has been the focus of the preceding sections. Two objects are Spherepop-identical when their normal forms are equal as event words:

$$X \equiv_{\mathrm{sp}} Y \quad \Longleftrightarrow \quad (\mathrm{spherepop}(X))_{\mathrm{nf}} = (\mathrm{spherepop}(Y))_{\mathrm{nf}}.$$

A strictly coarser relation is *causal isomorphism*, which identifies objects when their ancestral event graphs are isomorphic as labeled directed graphs but does not require

that the vertex sets be literally equal:

$$X \equiv_{\text{ci}} Y \quad \Longleftrightarrow \quad G_X \cong G_Y.$$

Under causal isomorphism two objects are equivalent when they arose from histories with the same causal shape and the same event types at each node, even if the specific identifiers of intermediate regions differ. This relation is useful in contexts where the labels of intermediate states are considered implementation details rather than constitutive features of identity.

Coarser still is *trace equivalence*, which identifies objects when their sets of possible executions—their sets of linear extensions of the causal poset—coincide:

$$X \equiv_{\text{tr}} Y \quad \Longleftrightarrow \quad \text{Lin}(G_X) = \text{Lin}(G_Y),$$

where $\text{Lin}(G)$ denotes the set of all linear extensions of the event graph $G$. Trace equivalence captures the idea that two objects are indistinguishable with respect to every possible sequential observation of their histories, while still retaining causal information beyond what is present in the final configuration.

The coarsest relation in this family is *structural isomorphism*, which identifies objects when their present configurations are isomorphic without any reference to history:

$$X \equiv_{\text{struct}} Y \quad \Longleftrightarrow \quad C_X \cong C_Y,$$

where $C_X$ denotes the present configuration of $X$. This is the set- theoretic notion of extensional identity discussed in Section 15, which discards all historical information.

These four relations are ordered by inclusion:

$$\equiv_{\text{sp}} \subseteq \equiv_{\text{ci}} \subseteq \equiv_{\text{tr}} \subseteq \equiv_{\text{struct}} .$$

Each successive relation is strictly coarser in general, though specific examples may make some pairs coincide. For instance, in a system with no branching, all linear event words with the same length over the same alphabet are already trace-equivalent, so $\equiv_{\text{sp}}$ and $\equiv_{\text{tr}}$ coincide.

The choice of which equivalence relation to employ depends on the application. For auditing and accountability, full Spherepop identity is appropriate. For type-theoretic reasoning about program behavior, causal isomorphism may suffice. For behavioral equivalences in process calculi, trace equivalence connects naturally to the Mazurkiewicz tradition [Mazurkiewicz, 1987]. For extensional mathematics, structural isomorphism suffices. The Spherepop framework provides the vocabulary to make these distinctions explicit and to reason about what is preserved or lost under each.

# 17 Spherepop as a Free Traced Monoidal Construction

The compositional structure of Spherepop histories places the framework naturally within the theory of categorical process theory. The primitive operations of the

formalism—the sequential composition of events, the independent parallel occurrence of events, and the merging and splitting constructions—correspond to the standard generators of a symmetric monoidal category. When the additional structure of internal closure is taken into account, the framework extends to a traced monoidal category. This section makes these connections explicit and uses them to clarify the universal properties of the Spherepop construction.

Let $\Sigma$ be a generating signature consisting of primitive event types. Each event type in $\Sigma$ is specified by the interface it consumes and the interface it produces, where an interface is a collection of regions that the event acts upon. From $\Sigma$ one constructs the free symmetric monoidal category $\mathcal{F}(\Sigma)$ in which morphisms correspond to event histories assembled from primitive events by sequential composition and monoidal product.

Sequential composition of events corresponds to categorical composition: if $E_1 : A \to B$ and $E_2 : B \to C$ are event types, then their sequential occurrence produces the composite morphism $E_2 \circ E_1 : A \to C$. The monoidal product $E_1 \otimes E_2 : A_1 \otimes A_2 \to B_1 \otimes B_2$ corresponds to the independent parallel occurrence of $E_1$ and $E_2$ acting on disjoint interfaces. An event word

$$(E_{i_1}, E_{i_2}, \ldots, E_{i_k})$$

therefore corresponds to the composite morphism

$$E_{i_k} \circ \cdots \circ E_{i_2} \circ E_{i_1}$$

in $\mathcal{F}(\Sigma)$. Branching and merging events, when represented as string diagrams in $\mathcal{F}(\Sigma)$, appear as boxes with multiple output wires and multiple input wires respectively, following the standard graphical calculus for monoidal categories [Baez and Stay, 2010].

The commutation rules of Section 7, which identify event words that differ only by the ordering of independent events, correspond precisely to the symmetry isomorphisms of the monoidal structure. In the free symmetric monoidal category, the symmetry natural transformation

$$\sigma_{A,B} : A \otimes B \xrightarrow{\sim} B \otimes A$$

witnesses the equivalence between the two orderings of independent events. The Spherepop normal form selects a canonical representative from each such equivalence class, mirroring the canonical form of morphisms in the free symmetric monoidal category modulo the coherence axioms.

Many Spherepop histories, however, involve intermediate regions that are created in one part of the history and subsequently consumed by a later merge event. Such patterns give rise to internal wires in the string diagram representation: wires that connect an output of one box to an input of a later box without being visible at the external boundary of the entire diagram. When such internal wires are systematically hidden, the appropriate categorical structure is a *traced monoidal category* [Baez and Stay, 2010].

In a traced symmetric monoidal category, the trace operation

$$\mathrm{Tr}^U_{A,B} : \mathrm{Hom}(A \otimes U,\, B \otimes U) \to \mathrm{Hom}(A, B)$$

formalizes the process of routing an intermediate wire of type $U$ from one stage of a diagram back into a later stage while hiding it from the external interface. The resulting morphism from $A$ to $B$ retains only the observable boundary behavior, abstracting over the internal routing.

Spherepop merge events, and more generally any situation in which intermediate regions are created and subsequently absorbed, can be interpreted as trace operations applied to the underlying string diagram. A history in which a region $U$ is produced by one event and consumed by a later merge event contributes a traced component to the diagram, even though $U$ may not appear in the final configuration.

The Spherepop framework may therefore be interpreted as the free traced symmetric monoidal category $\mathcal{F}_{\mathrm{tr}}(\Sigma)$ generated by the event signature $\Sigma$. The universal property of this construction states that any assignment of concrete morphisms to the primitive event types, respecting the traced monoidal structure of the target category, extends uniquely to a traced monoidal functor from $\mathcal{F}_{\mathrm{tr}}(\Sigma)$. Every Spherepop realization functor of Section 8 is an instance of this universal property.

What distinguishes Spherepop from an abstract categorical formalism is its ontological interpretation. In an ordinary traced monoidal category, morphisms represent processes and objects represent interfaces. In Spherepop, the morphisms are not merely descriptions of processes but constitute the identities of the objects they produce. A Spherepop object is an output boundary *together with* its generating morphism class in the free traced monoidal category. Two output boundaries are the same Spherepop object if and only if they are generated by the same morphism.

This identification of objects with their generating morphisms is the categorical formulation of the historical identity criterion developed throughout the paper. It situates the Spherepop ontology within the established framework of categorical process theory while emphasizing the philosophical commitment that distinguishes it from purely operational formulations of that theory.

# 18 Provenance Graphs and Distributed Event Logs

The Spherepop framework bears a close relationship to provenance tracking systems used in distributed computation and data management. In such systems the central problem is to record not merely the current state of a dataset but the sequence of operations that produced it. The resulting provenance graph serves as a historical record that allows users to verify, reproduce, and audit the transformations that generated a given artifact. This section examines the connection between Spherepop and provenance systems, distributed event logs, and blockchain-style ledgers, arguing that Spherepop provides a mathematical abstraction of patterns that already appear in practical systems.

A provenance graph typically consists of three categories of entities: data objects, transformation processes, and dependency relations [Spivak, 2014]. Processes consume input objects and produce output objects, and the directed edges of the graph record these causal dependencies. The structure that results is a directed acyclic graph representing the lineage of every object in the system. Spherepop histories may be interpreted as a formal abstraction of exactly this structure: regions correspond to data objects, events correspond to transformation processes, and the ancestral subgraph $G_X$ defined in Section 6 is precisely the provenance graph of the object $X$.

This interpretation makes precise a practical motivation for the Spherepop identity criterion. In many data-management contexts it is insufficient to know that two datasets are structurally identical; one must also know whether they were produced by the same sequence of transformations. Two files that contain identical data may nevertheless be considered distinct artifacts if they were produced by different computational workflows, because their different workflows may carry different validity guarantees, different institutional authorizations, or different reliability expectations. The Spherepop criterion captures this distinction exactly: two objects are identical only when their entire generative histories coincide.

Distributed event logs provide a closely related example. In modern distributed systems each component records a sequence of events representing local state transitions, and the global behavior of the system is reconstructed by combining these logs into a partial order of events, typically using logical clock mechanisms or causal message relationships. The structure that results is again a directed acyclic event graph, precisely of the form studied in Section 6. The causal poset of Section 11 formalizes the partial order that distributed systems practitioners construct from their logs, and the normalization procedure of Section 7 corresponds to the operation of computing a canonical linearization of concurrent events for purposes of comparison or replay.

Blockchain systems provide a particularly vivid illustration of the same principle. In a blockchain ledger the identity of a transaction is inseparable from its position within the chain of prior transactions. The ledger records an immutable append-only sequence of state transitions, and the validity of each entry depends on the entire chain of entries that preceded it. The identity of any ledger state is therefore constituted by the complete history of appended events, not by the final state alone. This is the Spherepop identity criterion applied to a specific computational substrate.

These examples suggest that Spherepop is not merely a theoretical construct but a mathematical articulation of patterns that arise independently in engineering practice. The framework provides a language for expressing precisely what these systems mean when they record a history, for reasoning about when two recorded histories represent the same event structure, and for studying the invariants that are preserved under different serialization and compression strategies. The normalization theory of Section 7 and the rewriting system of Appendix B apply directly: the question of whether two distributed logs represent the same global computation is precisely the question of whether their corresponding event words normalize to the same Spherepop normal form.

# 19 Conclusion

Spherepop develops a coherent and mathematically precise alternative to structurally or set-theoretically based criteria of identity. The central claim is that an object's identity is constituted by the irreversible sequence of events that produced it, formalized as the normal form $(\mathrm{spherepop}(X))_{\mathrm{nf}}$. This claim has been explored through a layered sequence of representations and theoretical developments: linear event words, directed acyclic event graphs, causal posets and their linear extensions, categorical diagrams of morphism compositions, realization functors from abstract causal histories into concrete systems, pushout-based accounts of merge events, geometric interpretations involving entropy landscapes and widening traces, a small-step operational semantics grounded in labeled transition systems, measures of historical complexity, the historical reconstruction problem and its irreversibility interpretation, a stratified family of equivalence relations ranging from full historical identity to structural isomorphism, a functorial characterization as a free traced monoidal construction, and connections to provenance systems, distributed event logs, and blockchain ledgers.

Each of these developments illuminates a different facet of the same underlying commitment. The event-word notation makes sequential historical structure explicit and directly comparable. The DAG and poset representations expose the parallel causal structure that any linear serialization necessarily obscures. The categorical and functorial formulations show how the identity criterion interacts with the standard infrastructure of composition, universality, and coherence. The operational semantics grounds the formalism in the standard vocabulary of process calculi, clarifying the relationship between Spherepop identity and bisimulation equivalences. The entropy landscape and complexity measures provide physical and combinatorial grounding respectively for the directionality and structural profile of histories. The stratified equivalence relations reveal that the strict Spherepop criterion occupies a specific position in a well-ordered lattice of identity notions, and the connection to provenance and distributed systems demonstrates that these notions have independent practical significance.

Together, these developments support a process ontology in which objects are terminal nodes of irreversible event structures rather than primitive persistent entities. Identity emerges from the structure of generative processes, and two objects that share a structural description at the present moment may nevertheless be genuinely distinct if they arrived at that description by different paths. The conclusion drawn is not merely that such distinctions can be made but that in a large and important class of systems they should be made: provenance, accountability, reproducibility, and thermodynamic irreversibility all depend on retaining the historical record that set-theoretic identity discards.

The framework connects naturally to several established traditions. The Mazurkiewicz trace theory [Mazurkiewicz, 1987] provides a concurrency-theoretic precedent for treating event sequences as equivalence classes modulo independent commutation. The Petri net formalism [Petri, 1966] provides an operational picture of branching and merging

processes. The categorical process theory initiated by Baez and Stay [2010] and developed by Spivak [2014] provides the language of functors, natural transformations, and universal constructions that gives the Spherepop semantics its rigorous backbone. The domain-theoretic tradition reviewed in Abramsky and Jung [1994] provides a precedent for process-based accounts of computational identity.

Several directions for further development are suggested by the present treatment. The normalization procedure of Section 7 and Appendix B should be examined for confluence and termination in the technical sense of abstract rewriting systems. The relationship between the weighted entropy measure $S(X)$ and concrete thermodynamic or information- theoretic quantities should be made precise through specific choices of the weight function $w$. The complexity measures of Section 12 should be organized into a systematic complexity theory of histories, with bounds on their interactions and characterizations of achievable complexity profiles. The free traced monoidal characterization of Section 17 should be sharpened into a precise universal property theorem with explicit statement of the adjunction involved. The connection to derived categories and sheaf-theoretic gluing conditions outlined in Appendix D should be developed into a precise embedding theorem. And the relationship to hypergraph categories and double categories of open systems, which the string diagram and merge-event constructions already suggest, should be explored as a possible further generalization beyond the traced monoidal framework. The present essay aims to establish the conceptual and formal foundations on which these more technical investigations can proceed.

# A A BNF Grammar for Spherepop Expressions

The abstract syntax of Spherepop expressions may be given precisely by a grammar in Backus–Naur Form (BNF). This grammar generates the symbolic representations of event histories that have been used throughout the main text, and provides a foundation for the rewriting system of Appendix B.

The lexical primitives are identifiers for regions and events. Identifiers are sequences of letters and digits beginning with a letter.

```
<identifier> ::= <letter>
               | <identifier> <letter>
               | <identifier> <digit>

<letter>     ::= a | b | ... | z | A | B | ... | Z
<digit>      ::= 0 | 1 | 2 | ... | 9
```

Event names are identifiers equipped with a numeric subscript, representing the label of the event in the global enumeration.

```
<event>      ::= "E" <digit-seq>
<digit-seq>  ::= <digit> | <digit> <digit-seq>
```

An event word records the ordered sequence of events in a history. The empty event word, corresponding to the identity object with no generative history, is denoted $\varepsilon$.

```
<event-word> ::= <epsilon>
               | <event>
               | <event> "," <event-word>
```

The Spherepop expression of an object wraps its event word in the canonical notation introduced in Section 3.

```
<spherepop-expr> ::= "(" "spherepop" "(" <object> ")" ")"
<object>         ::= <identifier>
```

Branching and merging events are represented by the following productions, which correspond to the split and merge constructions discussed in Sections 2 and 9.

```
<split>  ::= "split" "(" <object> ")" "->" <object> "," <object>
<merge>  ::= "merge" "(" <object> "," <object> ")" "->" <object>
```

The full syntactic description of a history is a sequence of statements, each of which is either a split, a merge, or a direct event transformation.

```
<history>         ::= <statement>
                    | <statement> <history>
<statement>       ::= <split>
                    | <merge>
                    | <event-transform>
<event-transform>::= <object> "->" <event> "->" <object>
```

A normalized Spherepop expression is one in which the event word has been ordered according to the causal dependencies of the event graph, as described in Section 7. The grammar for normal forms therefore coincides with the grammar for event words, with the additional requirement that the ordering respects the partial order induced by the event graph.

```
<normal-form> ::= "(" <event-word> ")"
```

Two Spherepop expressions represent identical objects precisely when they normalize to the same normal form according to the rewriting rules of Appendix B. The grammar therefore provides the syntactic substrate on which the normalization theory operates, and together they supply a decision procedure for Spherepop identity: parse the expression, construct the event graph, compute a topological ordering, and compare the resulting normal forms.

# B Rewriting Rules and Normalization

The normalization procedure described in Section 7 may be given a precise operational account in terms of a rewriting system acting on Spherepop expressions. Rewriting systems provide a standard framework for defining canonical forms through the iterated application of local rules, and their confluence and termination properties guarantee that the normalization procedure is well-defined and terminates [Abramsky and Jung, 1994].

The central rewriting rule exploits the independence relation $\parallel$ defined in Section 7. When two adjacent events in an event word are independent, their order may be exchanged.

$$(E_i, E_j) \longrightarrow (E_j, E_i) \quad \text{whenever } E_i \parallel E_j.$$

This rule is the Spherepop analogue of the commutation relations that define Mazurkiewicz trace equivalence [Mazurkiewicz, 1987]. By applying this rule repeatedly in a prescribed order—for instance, by always commuting the leftmost independent pair—one obtains the normal form of the history.

In addition to commutation, the rewriting system includes rules for restructuring split and merge expressions. A merge expression $\text{merge}(A, B) \to D$ may replace any pair of incoming edges into the vertex representing $D$ in the event graph, while a split expression

split$(R) \to A, B$ may replace a pair of outgoing edges from $R$. These structural rules convert compound symbolic descriptions of event graphs into standardized forms that are amenable to the commutation rule.

The normalization algorithm proceeds in three conceptual phases. In the first phase, the symbolic expression is parsed and the corresponding event graph is constructed explicitly. Vertices are created for each region appearing in the expression, and directed edges are inserted for each event. In the second phase, a topological ordering of the event graph is computed; this is always possible since the graph is a DAG. In the third phase, commutation rules are applied to bring the event word into agreement with the topological ordering.

The resulting sequence is the Spherepop normal form.

**Proposition 3** (Confluence)**.** *The commutation rewriting system is confluent: all rewriting paths from a given event word lead to the same normal form, regardless of the order in which commutation rules are applied.*

Confluence follows from the fact that the independence relation $\|$ is symmetric and the commutation rules satisfy the diamond property: whenever two distinct commutation rules are applicable to adjacent pairs in an event word, both can be applied in sequence, and the results join after a finite number of additional steps. The proof is analogous to the confluence proof for Mazurkiewicz traces [Mazurkiewicz, 1987] and will not be repeated here in full.

Confluence guarantees that the normal form is independent of the particular rewriting strategy used to compute it. Two Spherepop expressions represent the same historical object precisely when their rewriting sequences converge to the same normal form. The rewriting system thus provides an operational decision procedure for Spherepop identity that is both sound and complete with respect to the causal equivalence relation.

# C Causal Traces, Petri Nets, and String Diagram Representations

The event histories of Spherepop share structural features with three established mathematical frameworks for representing causal processes: trace theory, Petri nets, and string diagram formalisms. Examining these connections clarifies the position of Spherepop within the broader landscape of process-based formalisms and suggests directions for further technical development.

In the Mazurkiewicz trace theory of concurrency [Mazurkiewicz, 1987], a trace is an equivalence class of event sequences modulo the commutation of independent events. The event set $\mathcal{E}$ is equipped with an independence relation $\|$, and two sequences are identified when they are related by the exchange of adjacent independent events. The resulting equivalence classes—the traces—represent the causal content of an execution, abstracting away from the arbitrary serialization of concurrent actions. The Spherepop normal

form construction is precisely the operation of selecting a canonical representative from each trace class. The commutation rules of Appendix B are the Spherepop instantiation of the Mazurkiewicz commutation relations, and the confluence result established there corresponds to the standard confluence of the trace rewriting system.

The connection to Petri nets [Petri, 1966] is equally direct. In a Petri net, *places* represent conditions or resources and *transitions* represent events that consume and produce tokens at places. A simple correspondence maps Spherepop regions to places and Spherepop events to transitions. Under this mapping, split events correspond to transitions with multiple output places (producing a token in each of several descendant regions), while merge events correspond to transitions with multiple input places (requiring a token in each of the regions being combined before the merged region can be produced). A Spherepop history then corresponds to a *firing sequence* in the Petri net: a sequence of enabled transitions that produces the final configuration from the initial one. The DAG property of Spherepop event graphs corresponds to the property of the Petri net that the firing sequence contains no transition that could return the net to a previously visited marking.

String diagrams provide a geometric rather than algebraic language for compositional processes in monoidal categories [Baez and Stay, 2010]. In a string diagram, objects are represented by wires and morphisms are represented by boxes or nodes through which wires pass. Sequential composition is represented by connecting the output wire of one morphism to the input wire of the next; parallel composition is represented by placing wires side by side without connecting them. Spherepop events map naturally onto this language: a split event appears as a box with one input wire and two output wires, a merge event as a box with two input wires and one output wire, and a sequential transformation as a box with one input and one output. A complete Spherepop history becomes a compound string diagram assembled from these elementary components.

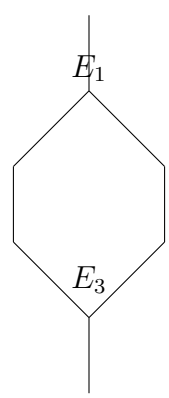


Figure 10: A string diagram representation of a split event $E_1$ followed by a merge event $E_3$. The single input wire at the top corresponds to the initial region $R$; the two intermediate wires carry the descendant regions $A$ and $B$; and the single output wire at the bottom corresponds to the merged region $D$.

As illustrated in Figure 10, string diagrams make the compositional structure of event histories visually transparent. Sequential composition corresponds to vertical stacking of diagram components, while parallel processes correspond to horizontal juxtaposition. The diagram in Figure 10 corresponds to the simple history $(\text{spherepop}(D)) = (E_1, E_3)$, where $E_1$ is the split and $E_3$ is the merge. More complex histories are represented by more elaborate compound diagrams.

What distinguishes Spherepop from the standard monoidal category framework is its emphasis on *historical identity* as the primary ontological criterion. The string diagrams and Petri net representations provide operational and geometric pictures of the same underlying structure, while the event-word and normal-form apparatus provides a canonical symbolic encoding. Spherepop integrates these perspectives into a unified framework where causal structure, compositional irreversibility, and identity under historical comparison are treated as equally fundamental features.

# D Spherepop and Derived Causal Categories

The constructions developed in the main text suggest that Spherepop histories may be situated within a higher-categorical framework in which the causal categories of Section 8 are enriched by the rewriting operations of Appendix B. This section sketches the relevant structures and their relationship to derived categories and sheaf theory, with the aim of indicating where more technical development could be pursued.

In the causal history category $\mathcal{H}$ introduced in Section 8, morphisms are irreversible events and their compositions are directed causal chains. The rewriting rules of Appendix B provide natural transformations between parallel event chains that differ only by the ordering of independent events. A commutation

$$(E_i, E_j) \Rightarrow (E_j, E_i) \quad \text{when } E_i \parallel E_j$$

may be interpreted as a 2-morphism in a bicategory or 2-category in which 1-morphisms are event chains and 2-morphisms are commutation steps. The confluence result of Appendix B then corresponds to the coherence of these 2-morphisms: any two sequences of commutation steps relating the same pair of event chains are connected by higher-order equivalences.

This higher-categorical structure suggests that the objects of Spherepop might profitably be studied as objects of a derived category, where the derived functor construction extracts invariant information from equivalence classes of resolutions. In the Spherepop context, the relevant resolutions are the various syntactic descriptions of a single event graph, and the derived invariant is the normal form. The normalization functor of Section 7 plays the role of the derived functor, extracting the canonical representative from a class of equivalent descriptions.

Large histories may also be assembled from smaller components defined on overlapping subgraphs of the event graph. If $\{H_i\}$ is a collection of local histories defined on overlapping regions and these histories agree on their pairwise intersections—that

is, the restrictions of $H_i$ and $H_j$ to the subgraph $G_i \cap G_j$ are equal—then a global history may be assembled by gluing. This gluing condition is precisely the condition that appears in sheaf theory when local sections over open sets of a topological space are assembled into a global section. Spherepop histories therefore exhibit the structure of a sheaf over the underlying causal graph, where the "open sets" are the hereditary sub-DAGs and the "sections" are compatible local histories.

The sheaf-like structure has a natural operational interpretation. In a distributed computation, different processors may have direct knowledge only of the events that occurred within their local scope. The global history of the computation is assembled from these local records by verifying consistency on the overlaps and then gluing compatible pieces together. The Spherepop sheaf structure captures exactly this assembly process, and the normal form provides a canonical global section that encodes the complete causal history of the computation.

From this perspective, Spherepop may be embedded into a broader framework of derived causal categories in which objects correspond to terminal nodes of event diagrams, morphisms correspond to irreversible processes, 2-morphisms represent rewriting equivalences between histories, and normalization extracts canonical representatives from these equivalence classes. The present essay has laid the conceptual and formal foundations on which this programme of technical development can proceed.

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